Ammonia

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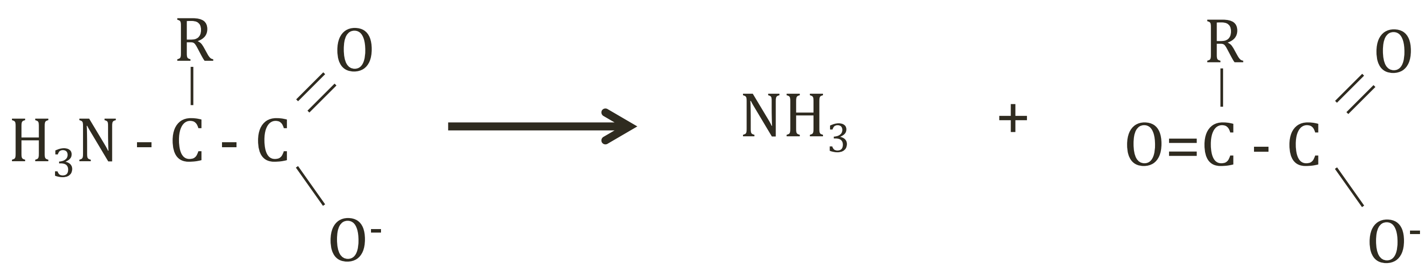
Amino Acid Breakdown

- No storage form of amino acids
- Unused amino acids broken down
- Amino group removed → NH_3 + α-keto acid



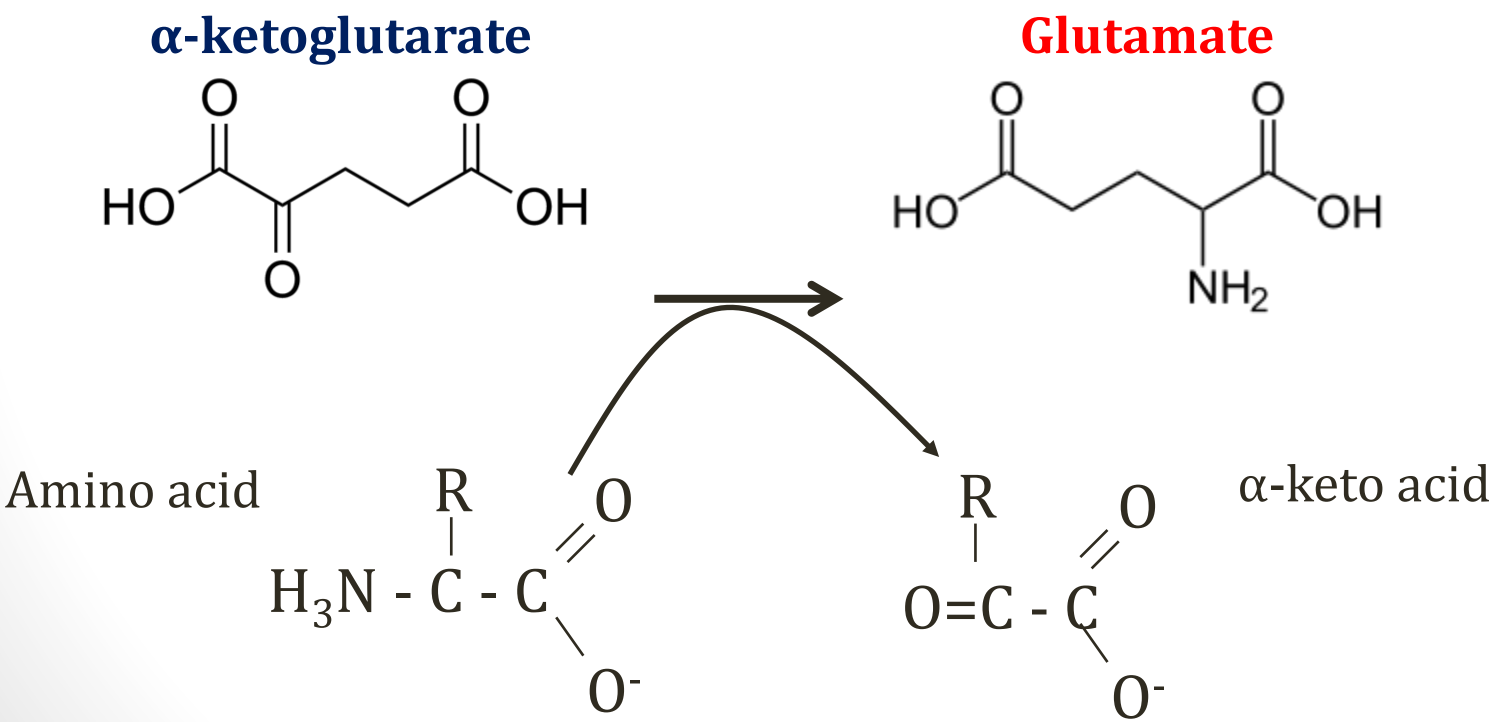
Ammonia

- Toxic to body
- Transferred to liver in a non-toxic structure
- Converted by liver to urea (non-toxic) for excretion

Pixabay/Public Domain

Amino Acid Breakdown

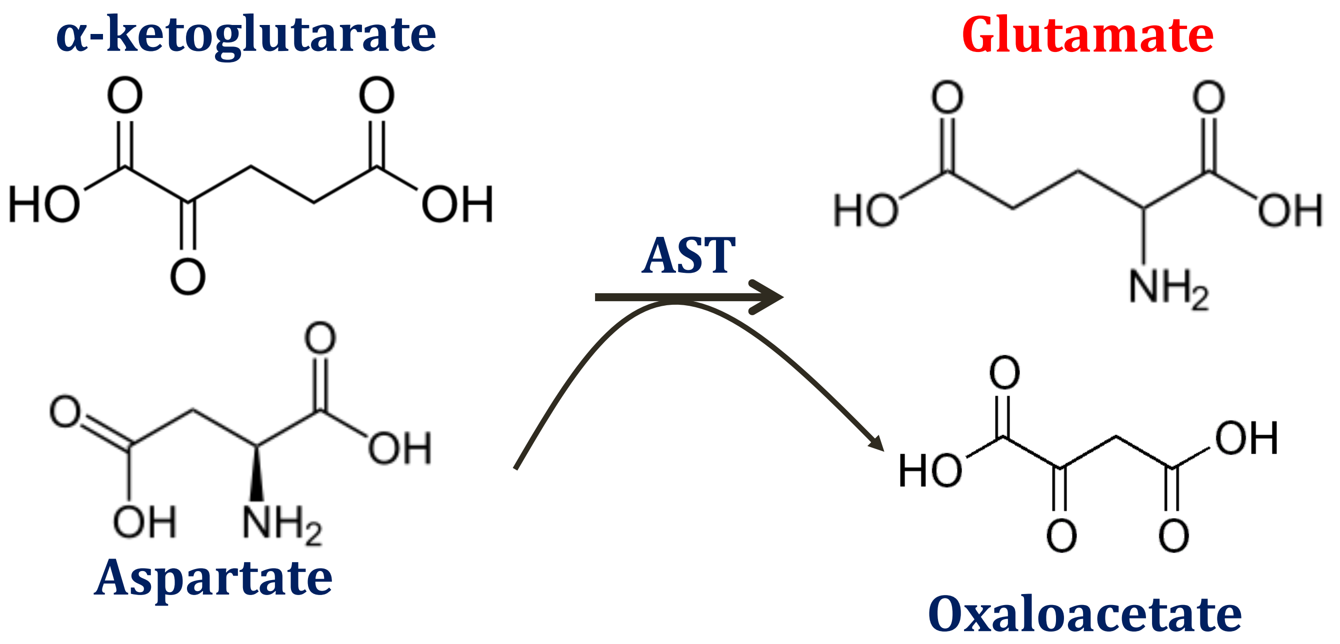
- Usual 1st step: removal of nitrogen by **transamination**
- Amino group passed to **glutamate**



Aminotransferases

- Transfer nitrogen from amino acids to glutamate
- All require **vitamin B6** as cofactor
- Two used as liver function tests:
  - Alanine aminotransferase (ALT)
  - Aspartate aminotransferase (AST)

Aminotransferases

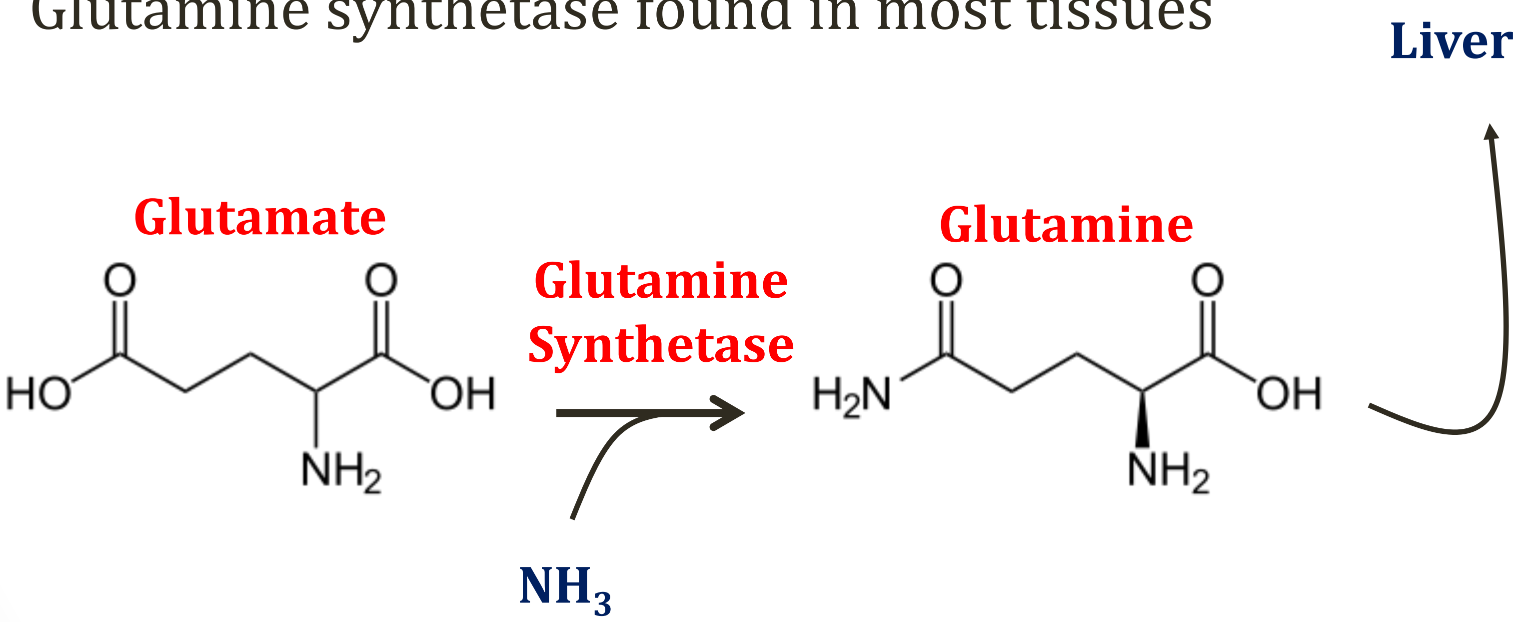


Glutamate

- Two methods for transfer of nitrogen from glutamate to liver for excretion in urea cycle
- #1: Glutamine synthesis
- #2: Alanine cycle

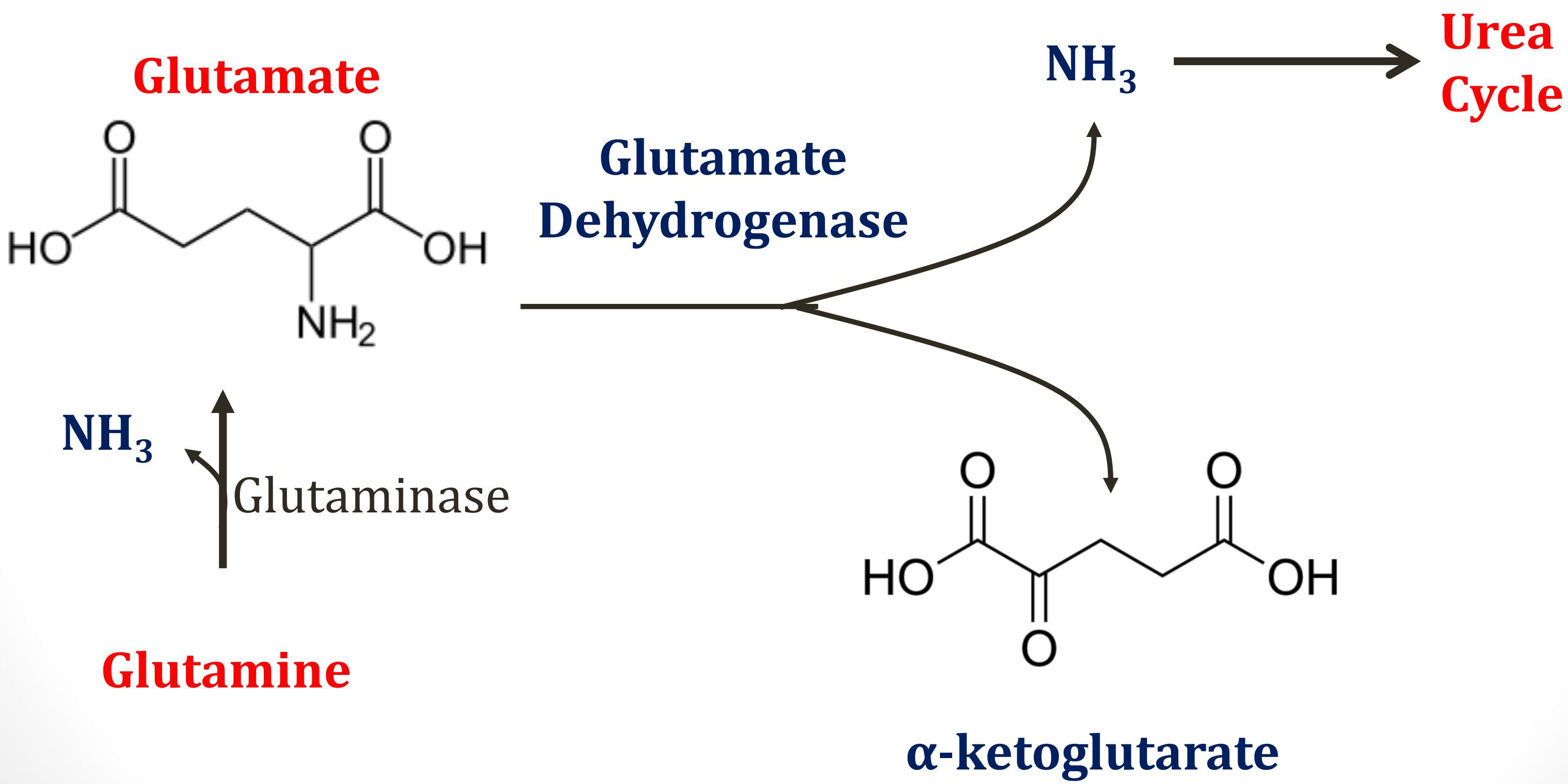
Glutamine

- Non-toxic
- Transfers nitrogen to liver for excretion
- Glutamine synthetase found in most tissues



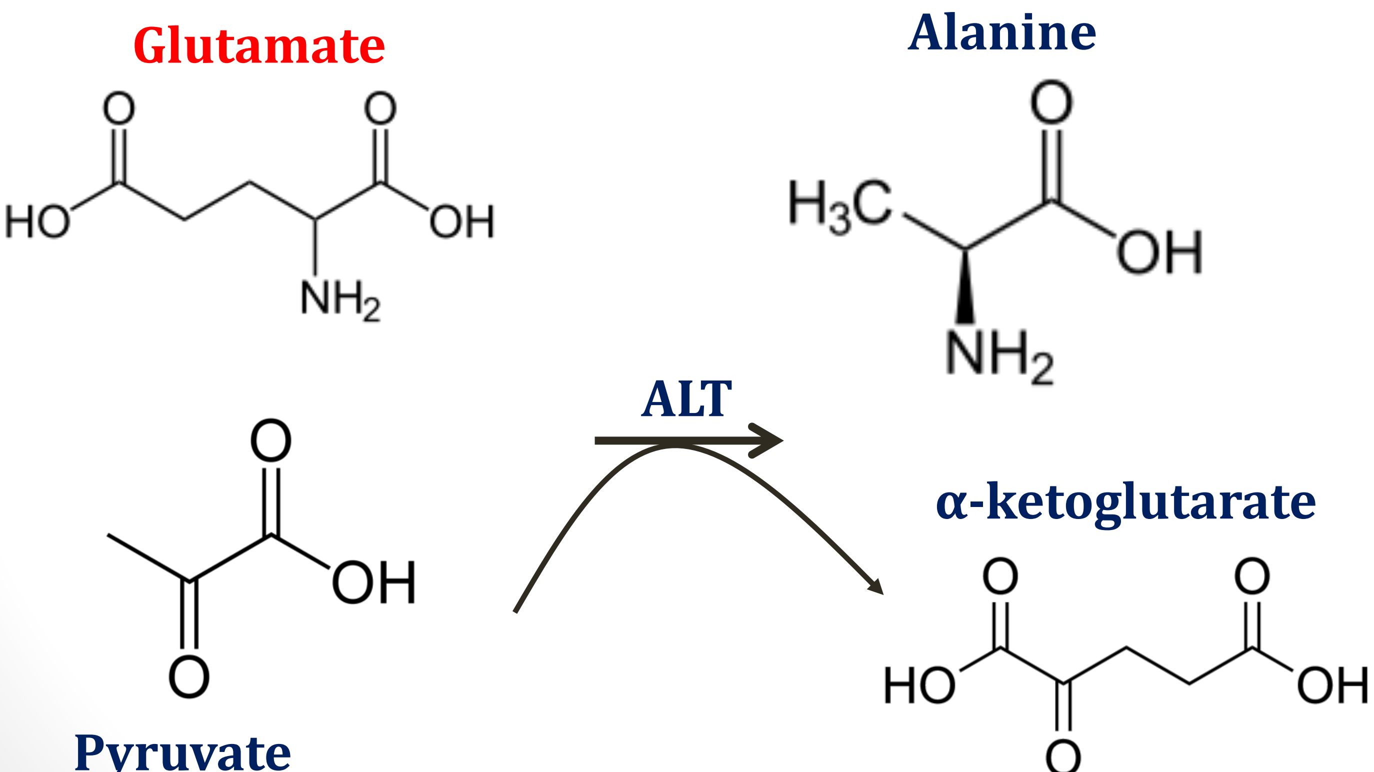
Glutamine

- In liver, glutamine converted back to glutamate



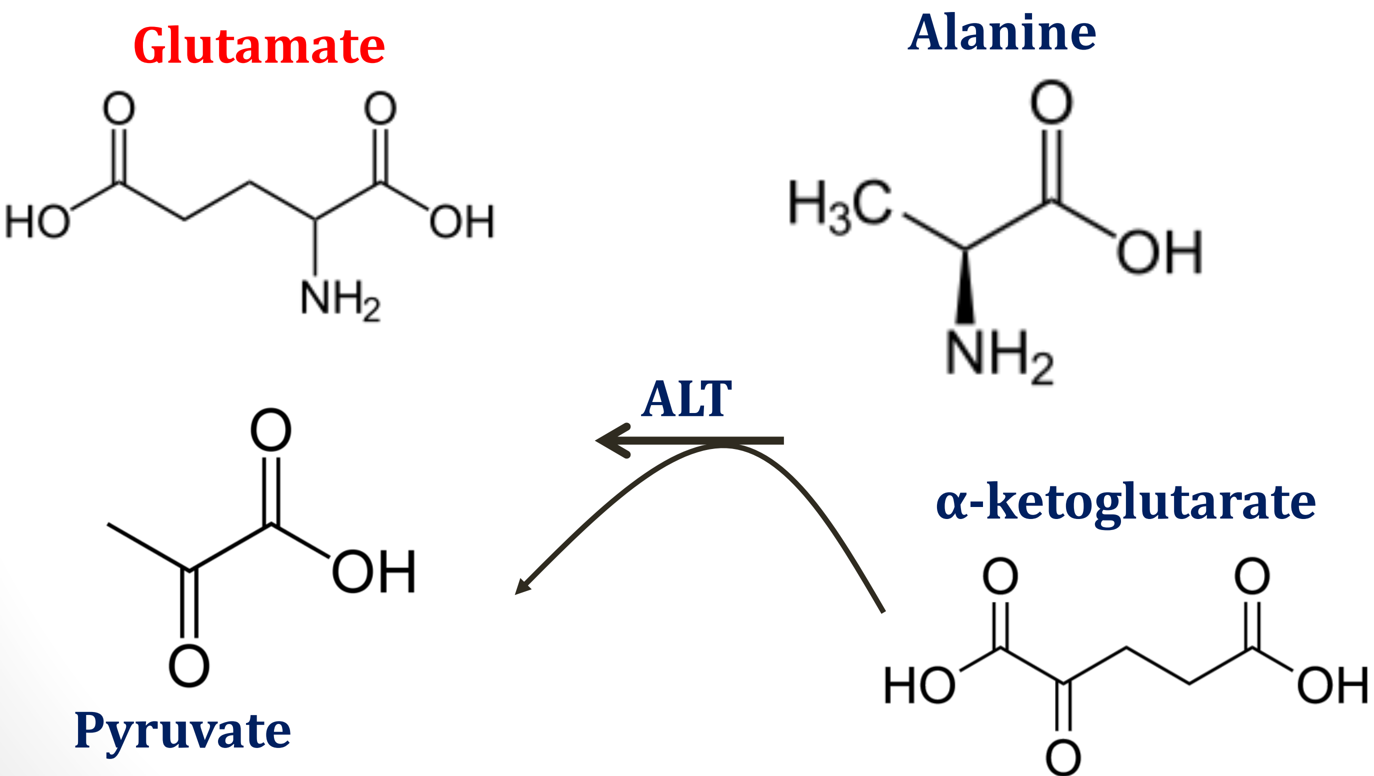
Alanine Cycle

- Used by **muscles** to transfer nitrogen to liver
- Glutamate nitrogen → alanine



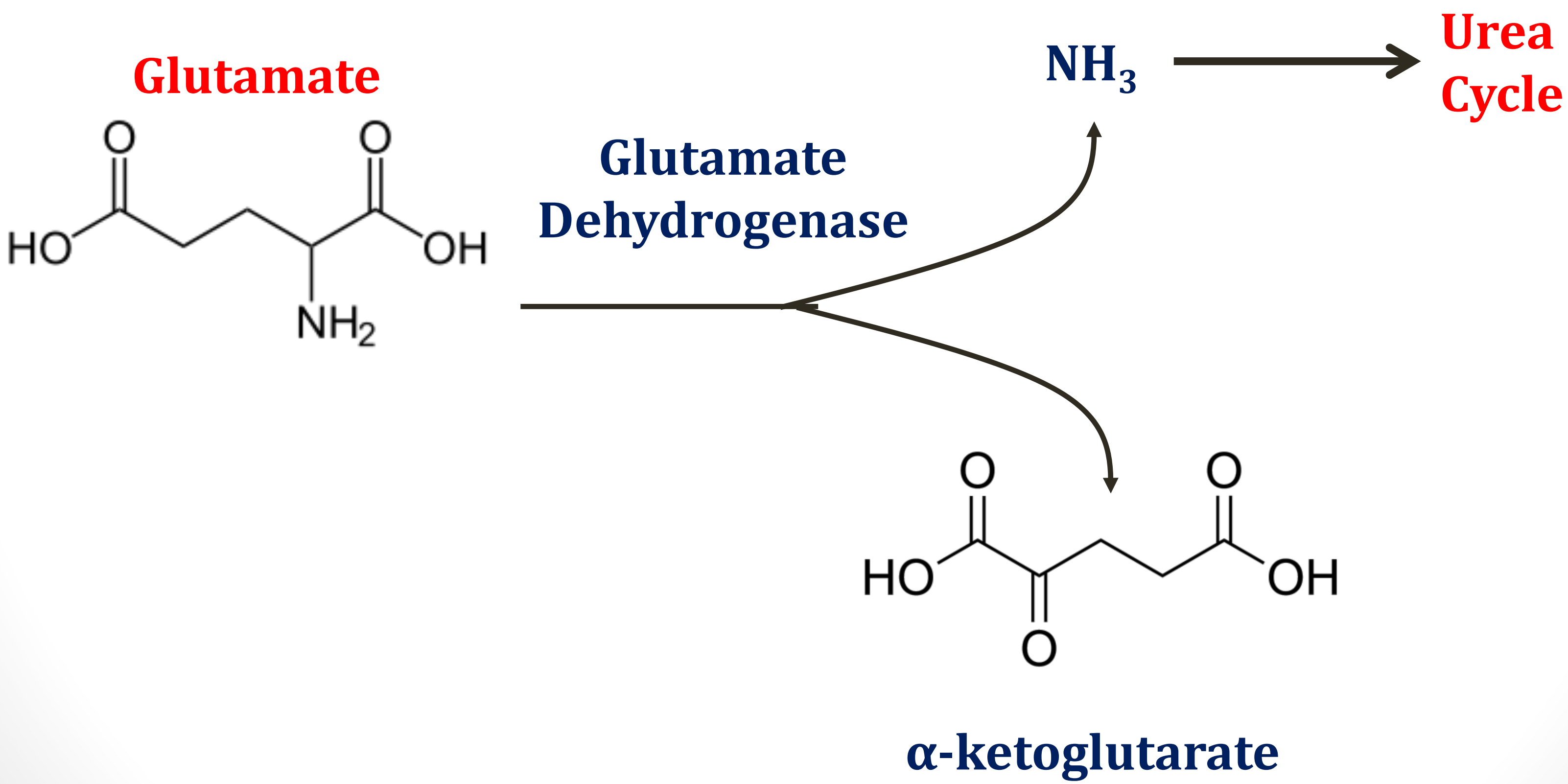
Alanine Cycle

- Alanine to liver → pyruvate
- Nitrogen transferred back to glutamate

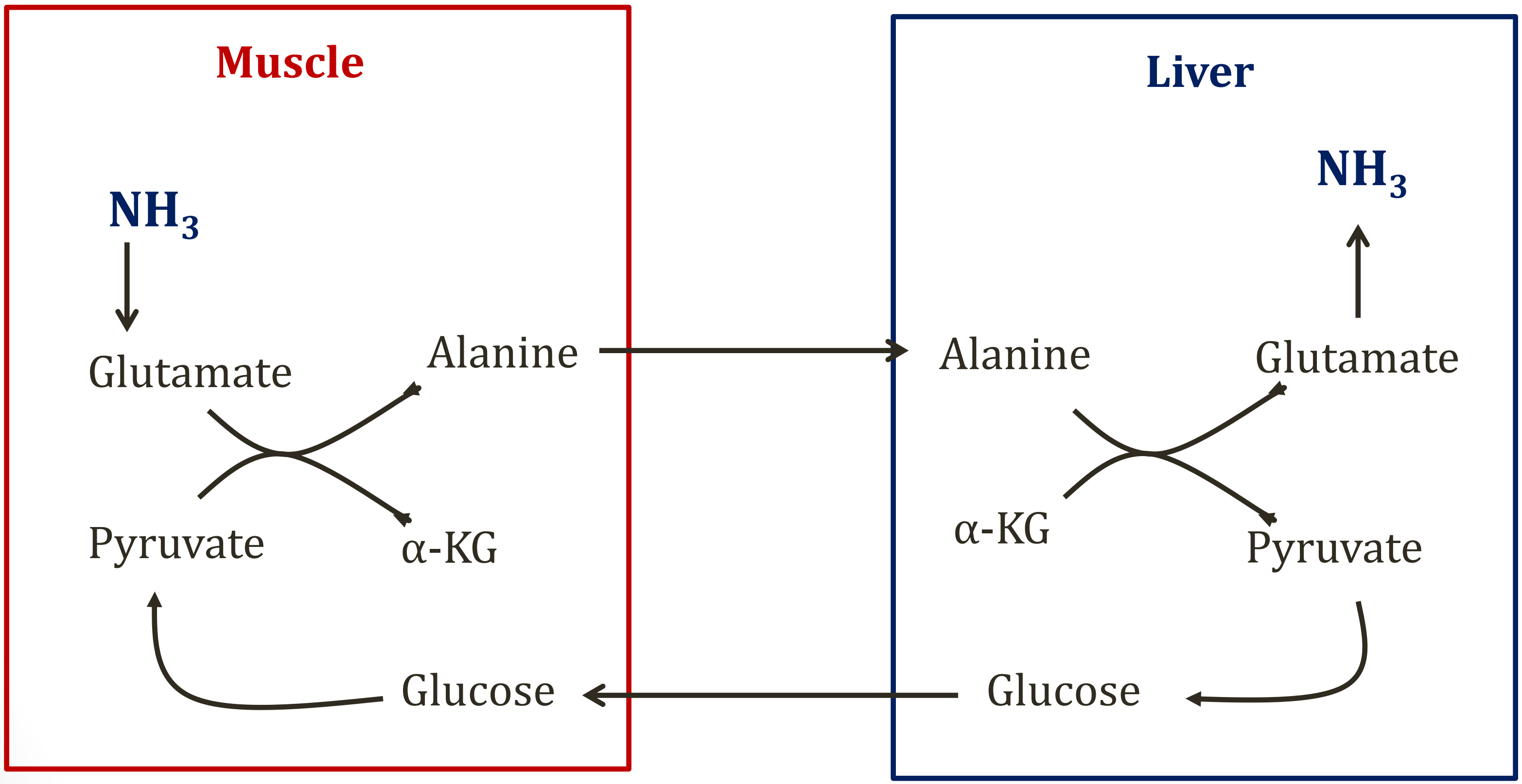


Alanine Cycle

- Nitrogen removed from glutamate



Alanine Cycle



Mitochondrial Disorders

- Inborn errors of metabolism
- Often deficient metabolism of pyruvate
  - Pyruvate carboxylase deficiency
  - Pyruvate dehydrogenase deficiency
- Elevated **alanine** and lactate

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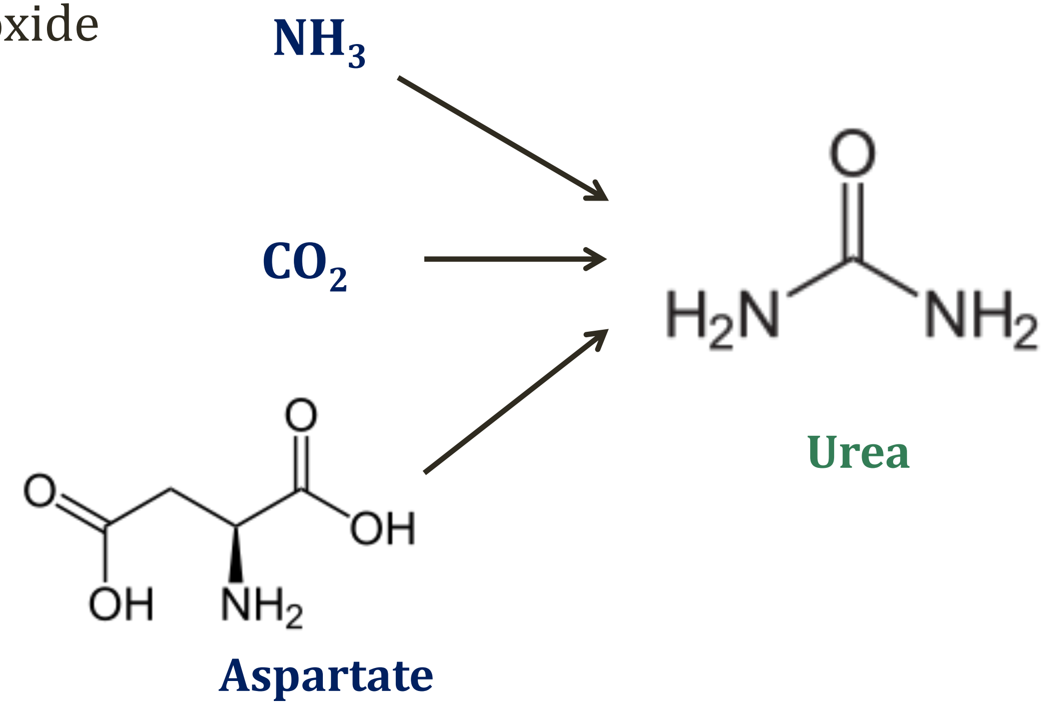
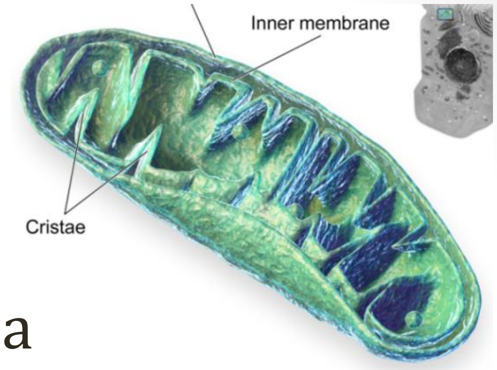
Urea Cycle

- Ammonia (NH_4^+) → Urea → Excretion in urine
- Urea synthesized from:
  - Ammonia
  - Carbon dioxide
  - Aspartate

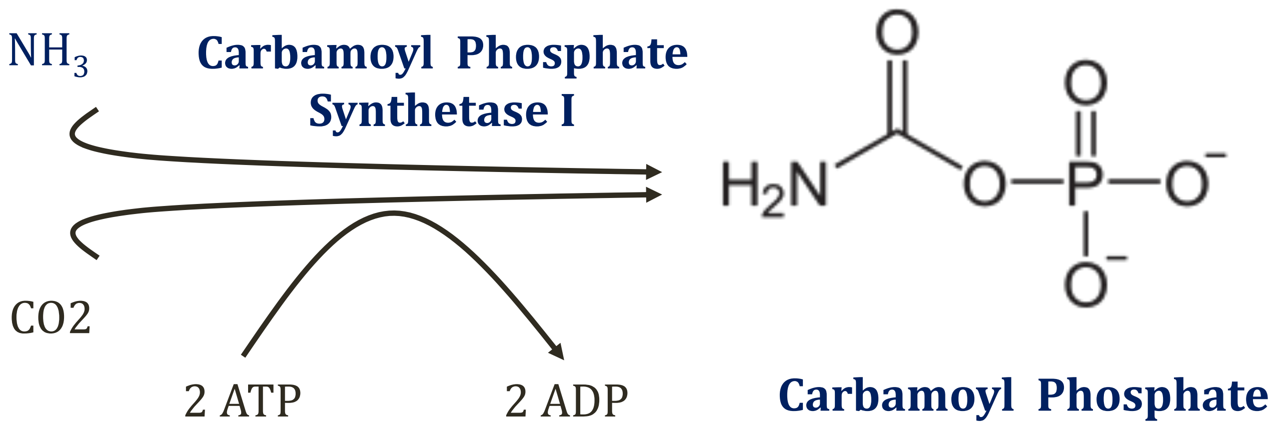


Urea Cycle



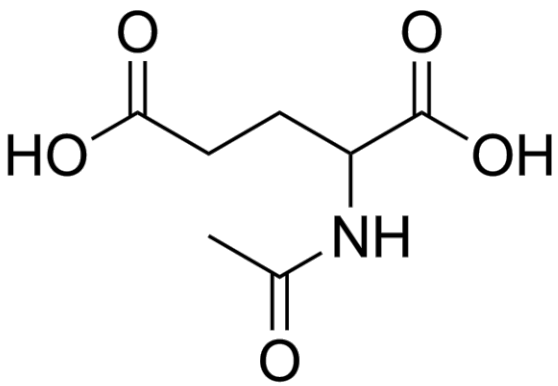
- First reaction (and 2nd) in mitochondria
- Rate limiting step

Blausen gallery 2014".
Wikiversity Journal of Medicine



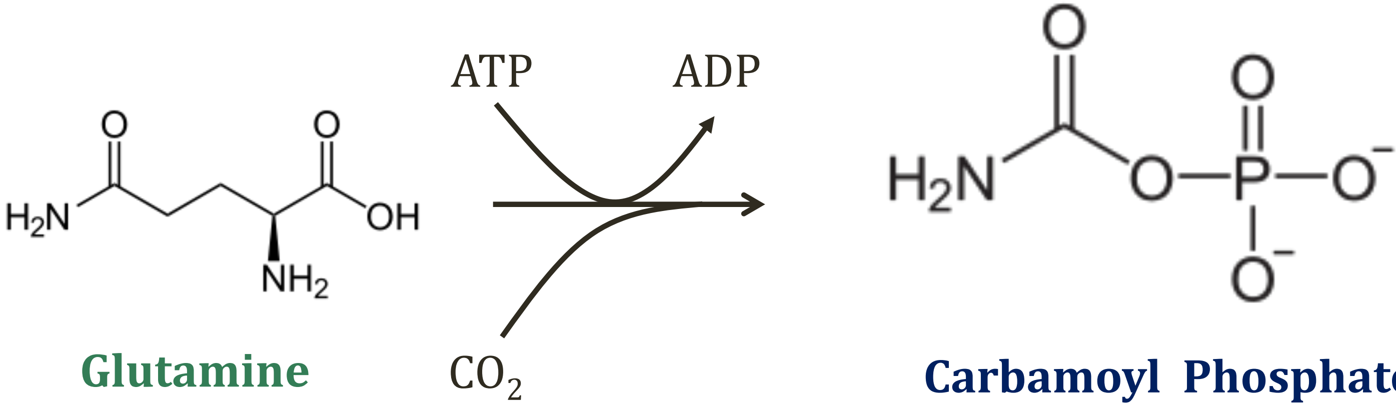
N-acetylglutamate

- Allosteric **activator**
- **Carbamoyl Phosphate Synthetase I**
- Enzyme will not function without this cofactor
- Synthesized from glutamate and acetyl CoA
- ↑ protein (fed state) → ↑ N-acetylglutamate
- Used to regulate urea cycle



Pyrimidine Synthesis

- **Carbamoyl phosphate synthetase II**

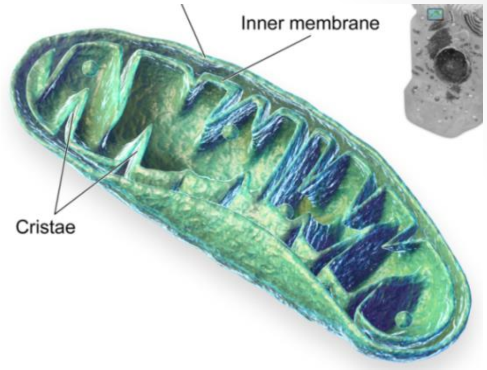


Glutamine

Carbamoyl phosphate synthetase II

Carbamoyl Phosphate

Urea Cycle



Blausen gallery 2014".
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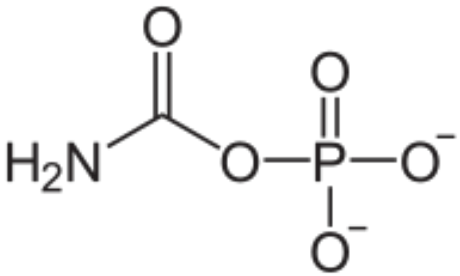
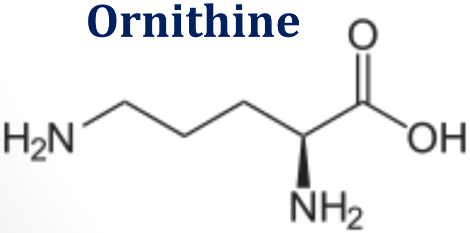
- Second reaction also in mitochondria



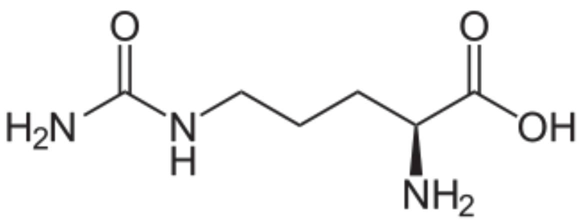
Carbamoyl Phosphate

+

Ornithine

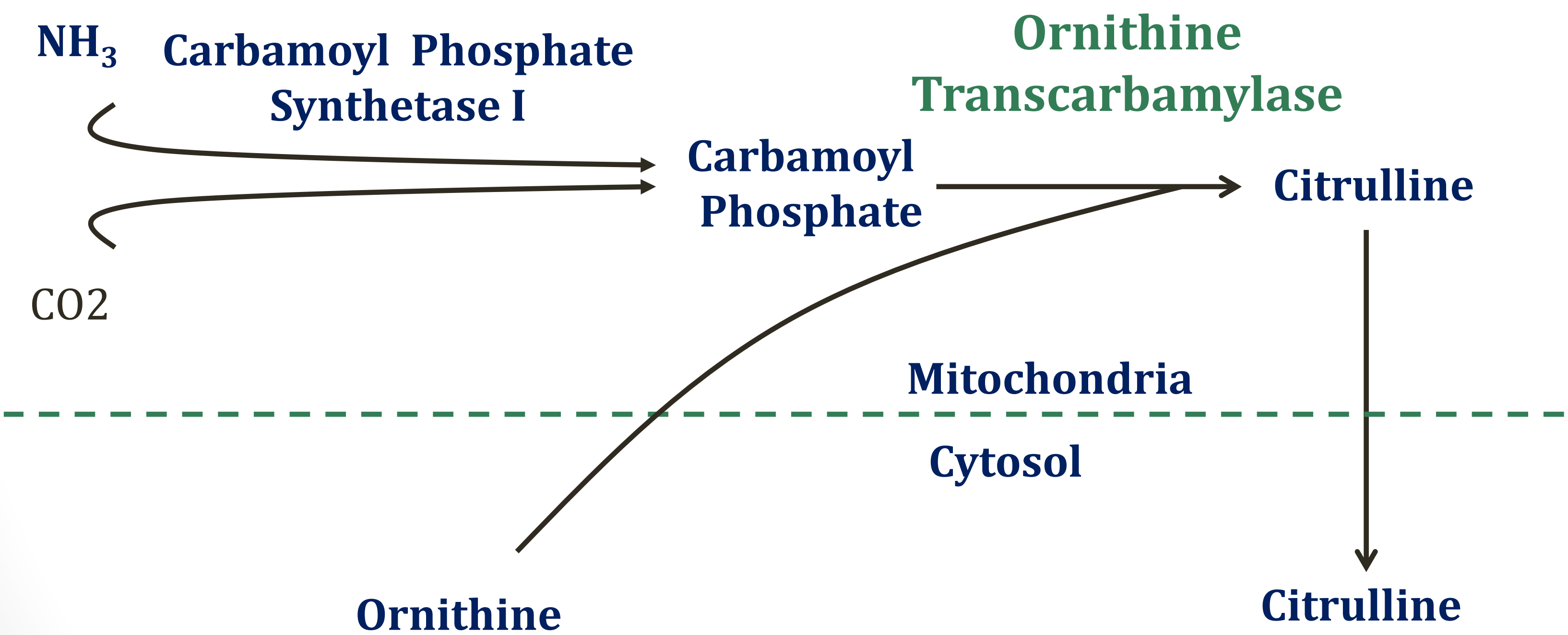


Ornithine Transcarbamylase



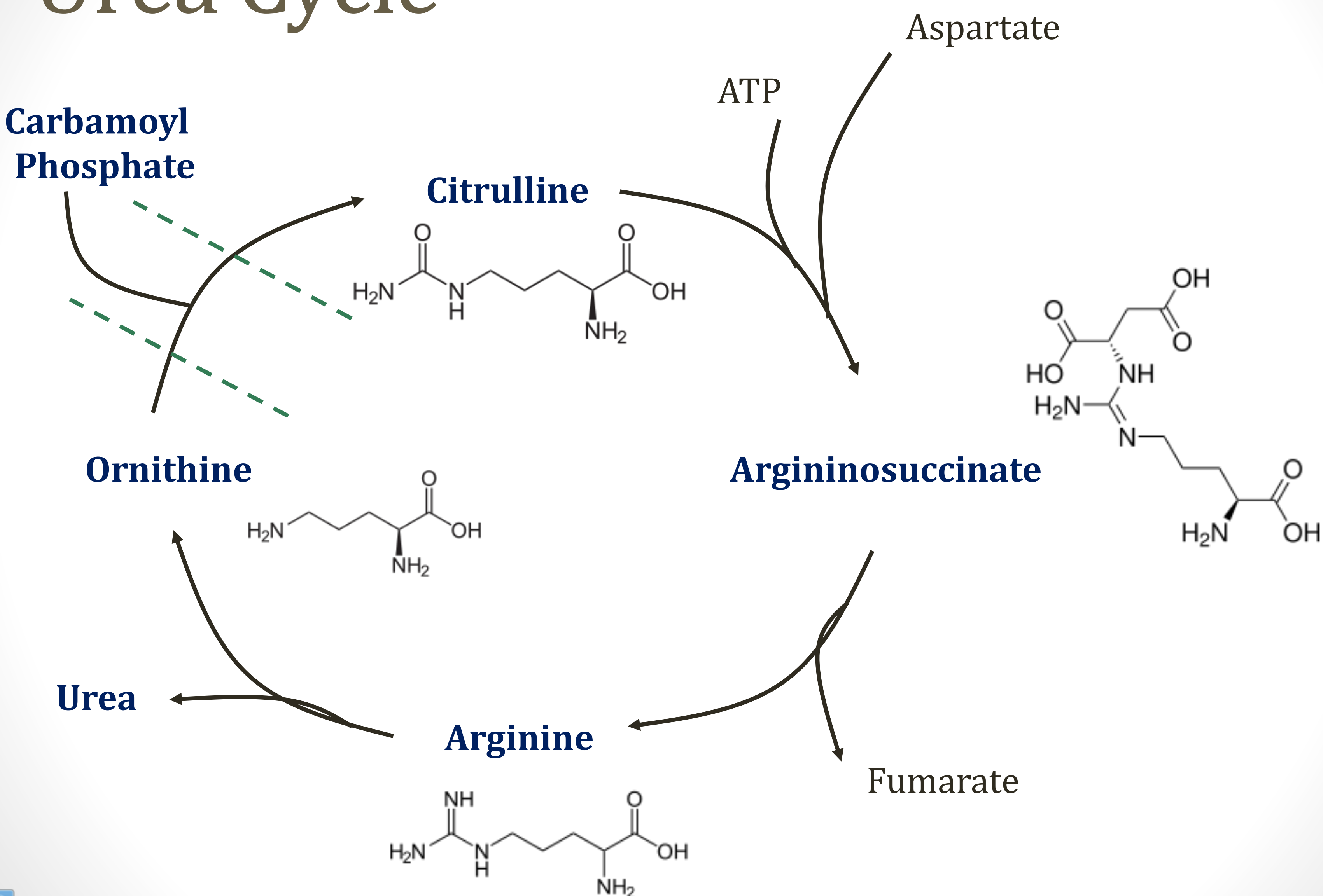
Citrulline

Urea Cycle



NH_3
Carbamoyl Phosphate Synthetase I
CO2
Ornithine Transcarbamylase
Carbamoyl Phosphate
Citrulline
Mitochondria
Cytosol
Ornithine
Citrulline

Urea Cycle


Aspartate
ATP
Carbamoyl Phosphate
Citrulline
Ornithine
Argininosuccinate
Urea
Arginine
Fumarate

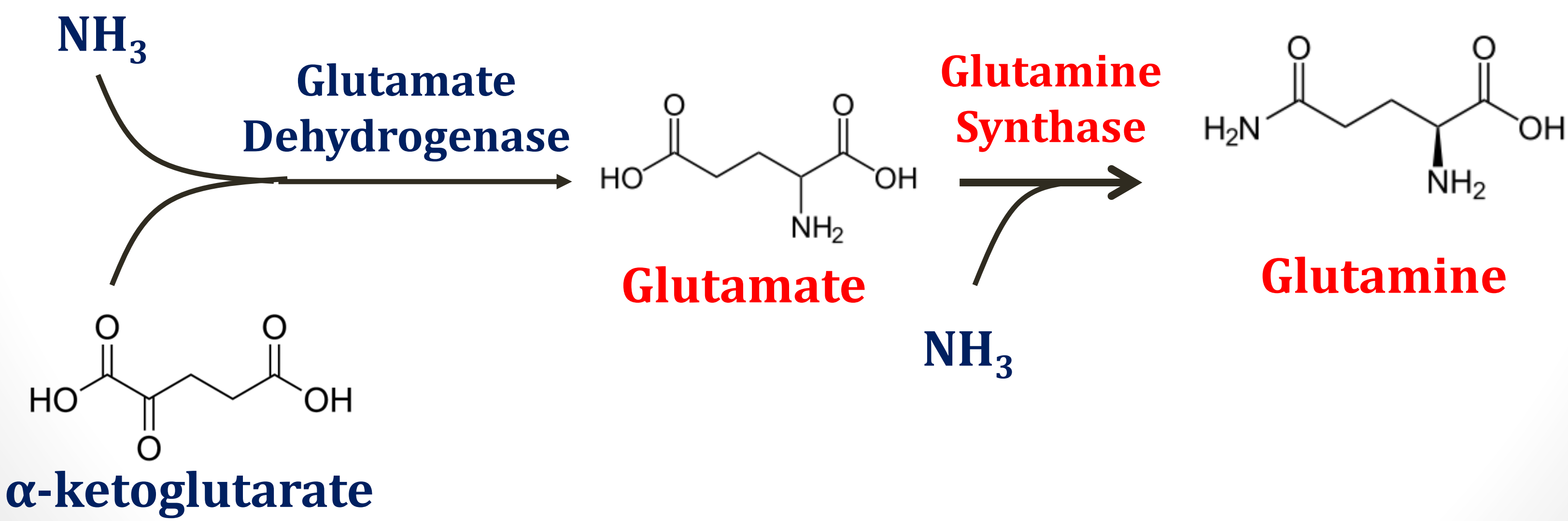
Citrulline

- Non-standard amino acid - not encoded by genome
- Incorporated into proteins via post-translational modification
- More incorporation in inflammation
- Anti-citrulline antibodies used in rheumatoid arthritis
    - Anti-cyclic citrullinated peptide antibodies (anti-CCP)
    - Up to 80% of patients with RA

James Heilman, MD/Wikipedia

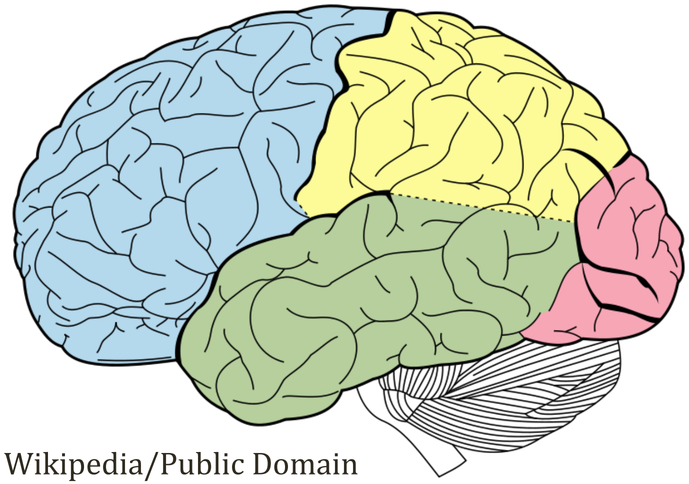
Hyperammonemia

- Results from any disruption urea cycle
  - Commonly seen in advanced liver disease
  - Rare cause: urea cycle disorders
- ↑ ammonia can deplete **α-ketoglutarate** (TCA cycle)



Hyperammonemia
Symptoms

- Main effect on **CNS**
- Can lead to cerebral edema
- Tremor (asterixis)
- Memory impairment
- Slurred speech
- Vomiting
- Can progress to coma



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Hyperammonemia

Treatment

~~STEAK~~

- Low protein diet
- Lactulose
  - Synthetic disaccharide (laxative)
  - Colon breakdown by bacteria to fatty acids
  - Lowers colonic pH; favors formation of NH_4+ over NH_3
  - NH_4^+ not absorbed → trapped in colon
  - Result: ↓plasma ammonia concentrations

Hyperammonemia

Treatment: Enzyme deficiencies only

- Ammonium Detoxicants
  - Sodium phenylbutyrate (oral)
  - Sodium phenylacetate-sodium benzoate (IV)
  - Conjugate with glutamine
  - Excreted in the urine → removal of nitrogen/ammonia
- Arginine supplementation
  - Urea cycle disorders make arginine essential

OTC Deficiency

Ornithine transcarbamylase deficiency

- Most common urea cycle disorder
- X linked recessive
- ↑ carbamoyl phosphate
- ↑ ammonia
- ↑ orotic acid (derived from carbamoyl phosphate)

Glutamine → **Carbamoyl Phosphate** → **Orotic Acid** → UMP → CMP / TMP

UMP Synthase
Bifunctional

Pyrimidine Synthesis

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OTC Deficiency

Ornithine transcarbamylase deficiency

- Presents in infancy or childhood
  - Depends on severity of defect
  - If severe, occurs after first several feedings (protein)
- Symptoms from hyperammonemia
- Somnolence, poor feeding
- Seizures
- Vomiting, lethargy, coma

OTC Deficiency

Ornithine transcarbamylase deficiency

- Don't confuse with orotic aciduria
  - Disorder of pyrimidine synthesis
  - Also has orotic aciduria
  - OTC only: ↑ **ammonia levels** (urea cycle dysfunction)
  - Ammonia → encephalopathy (child with lethargy, coma)

Citrullinemia

- Deficiency of **argininosuccinate synthase**
- Elevated **citrulline**
- Low arginine
- Hyperammonemia

Other Urea Cycle Disorders

- Deficiencies of each enzyme described
- All autosomal recessive except OTC deficiency
- All cause **hyperammonemia**
- Build-up of urea cycle intermediates